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# 🧪 Hands-On Lab: AWS Compute Optimizer for EC2 and EBS

## 🎯 Objective

Enable **AWS Compute Optimizer**, analyze your EC2/EBS resource usage, and review **cost-saving recommendations** to right-size infrastructure.

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## 🧱 Prerequisites

- AWS account with admin privileges
- At least one **running EC2 instance**
- Permissions:
  - `ComputeOptimizerFullAccess`
  - `CloudWatchReadOnlyAccess`
  - `ec2:DescribeInstances`

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## 🧩 Lab Structure

1. Enable AWS Compute Optimizer
2. Launch a test EC2 instance (if not already running)
3. Wait for data collection (minimum 12–24 hours)
4. View optimization recommendations

5. Analyze suggestions and take action

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## 🔧 Step 1: Enable AWS Compute Optimizer

1. Go to: [https://console.aws.amazon.com/compute-optimizer](https://console.aws.amazon.com/compute-optimizer)
2. If first time:
   - Click **"Get started"**
   - Choose:
     - ✅ **EC2 Instances**
     - ✅ **EBS Volumes**
     - ✅ **Auto Scaling Groups** *(optional)*
     - ✅ **Lambda Functions** *(optional)*
3. Click **Enable**

💡 **Note**: Data collection takes ~12–24 hours before recommendations show up.

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## 🖥️ Step 2: Launch a Test EC2 Instance (if needed)

If you don't already have EC2 running:

1. Go to **EC2 Console**
2. Launch an EC2 instance with:
   - AMI: Amazon Linux 2
   - Type: `t2.medium`

- Name: `ComputeOptimizerTest`
- Leave it running idle or under light load

Let it run for a **few hours to days** to gather usage data.

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## 🔍 Step 3: View Recommendations

1. Go to **AWS Compute Optimizer Console**
2. In the sidebar, select:
   - **EC2 Instance Recommendations**
   - **EBS Volume Recommendations**
3. For each resource, review:

| Field | Description |
|---|---|
| **Current Instance Type** | e.g., `t2.medium` |
| **Recommended Type** | e.g., `t3.small` |
| **Performance Risk** | Low, Medium, or High |
| **Savings Opportunity** | Monthly cost savings |
| **Utilization Metrics** | CPU, Memory, Network |

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## 🛠️ Step 4: Take Action (Manual Optimization)

If you agree with the recommendation:

### Option 1: Resize EC2 instance

1. Stop the instance

2. Change instance type (e.g., from `t2.medium` to `t3.small`)

3. Start the instance again

### Option 2: Change EBS Volume Type

1. Go to **EBS Console**

2. Modify volume → choose recommended volume type (e.g., gp2 → gp3)

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## 🧪 (Optional) Step 5: Use AWS CLI to Get Recommendations

aws compute-optimizer get-ec2-instance-recommendations \
  --region us-east-1 \
  --query "instanceRecommendations[*].{InstanceId:instanceId, Current:currentInstanceType, Recommend:recommendationOptions[0].instanceType, Risk:recommendationOptions[0].performanceRisk}"

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## 🧠 Summary

| Task | Completed |
|---|---|
| Enabled Compute Optimizer | ✅ |
| Launched EC2 (if needed) | ✅ |
| Waited for usage data | ✅ |
| Reviewed recommendations | ✅ |
| Took optimization action | ✅ |

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## ✅ Cleanup

- Stop or terminate the test EC2 instance (if created for the lab)
- Modify EBS volumes back if needed

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